Nafis Karim

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SUMMARY

CS PhD student at UIC and co-first author of AgentShield, accepted to IEEE S&P (Oakland, CORE A*) in my first PhD year. Research focuses on LLM agent security, behavioral fingerprinting, adversarial evaluation, and transformer-based information extraction. Skilled in Python, C/C++, Java, JavaScript/TypeScript, PyTorch, Hugging Face, Linux, Docker, and network/systems programming.

EDUCATION

Ph.D. in Computer Science Aug 2025 – May 2030 (Expected)
University of Illinois Chicago (UIC)
Advisor: Prof. Jason Polakis
Activities: Teaching Assistant, CS 107 (Intro to Computing and Programming) and CS 109 (MATLAB for Engineers)

B.Sc. in Computer Science and Engineering May 2019 – Jun 2024
Bangladesh University of Engineering and Technology (BUET) CGPA: 3.81/4.00
Awards: Dean's List Scholarship

PEER-REVIEWED PUBLICATIONS

AgentShield: Fortifying Web Applications Against AI Agents and Agentic Browsers
Moiz Ali*, **Nafis Karim***, Rownak Ul Islam, Jyotirmay Chauhan, Dhruv Kucchal, Jason Polakis (*Joint first authors)
IEEE Symposium on Security and Privacy (S&P), 2027 (**CORE A***, **15.8% acceptance rate**)

- Evaluated 14 LLM agents (e.g., Perplexity Comet, ChatGPT Atlas, Claude for Chrome) against deployed anti-bot defenses (reCAPTCHA, Turnstile), revealing many fail to distinguish agents from humans.
- Built AgentShield, a behavioral fingerprinting framework that detects agents via dynamic forms and interaction telemetry; shipped it as npm packages with all code, prompts, and data released for reproducibility.
- Discovered Extinction-Induced Policy Drift, a novel attack that makes agents bypass their own safeguards.

ADDITIONAL RESEARCH EXPERIENCE

Undergraduate Thesis: Cyber Threat Intelligence Extraction with Transformer-based LLMs BUET
Supervisor: Dr. Md. Shohrab Hossain; Collaborators: Prof. Ren-Hung Hwang, Prof. Ying-Dar Lin (NYCU, Taiwan)

- Benchmarked fine-tuned BERT/RoBERTa against few-shot GPT-4 and Gemini for NER on DNRTI and APTNER.
- Fine-tuned SecureBERT set a new state of the art on DNRTI (F1 0.859, 95.6% accuracy) vs. GPT-4's best F1 of 0.705.

COVID-19 Case Forecasting with a Custom Deep Learning Architecture BUET
Best Student Poster, NSysS 2023

- Built a hybrid CNN-RNN forecaster on case, vaccination, and hospitalization data; beat baselines on US/UK data.

PROJECTS

HTTPS Interception Proxy | **Python, TLS, Sockets, Docker** | [GitHub Repository]
Wrote a from-scratch TLS-terminating MITM proxy that handles CONNECT and logs decrypted auth traffic as JSON.
DNS Data Exfiltration Lab | **Python, Scapy, Docker** | [GitHub Repository]
Built a covert channel encoding files into DNS query labels, reassembled via transaction IDs; analyzed pcaps.
ARP Spoofing Lab | **Python, Scapy, Docker, tcpdump** | [GitHub Repository]
Staged ARP cache poisoning on a Dockerized network to intercept traffic to a Flask authentication server.
Cryptanalysis and Password Cracking | **Python, hashlib** | [GitHub Repository]
Broke substitution ciphers via frequency analysis; cracked salted hashes with dictionary and leetspeak attacks.
sub-C Compiler | **C, x86 Assembly** | [GitHub Repository]
Built a compiler for a C subset with lexical, syntax, and semantic analysis and x86 assembly code generation.

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript, TypeScript, Java, Bash, SQL
Bot and Agent Defense: FingerprintJS, reCAPTCHA v2/v3, Cloudflare Turnstile and AI Crawl Control, Suricata (JA4)
Network Security: Scapy, tcpdump, Wireshark, Nmap, TLS interception, ARP spoofing, DNS tunneling
Web and Infrastructure: Linux, Docker Compose, Ansible, nginx, Cloudflare Workers, Node.js, React, Git
Machine Learning: PyTorch, Hugging Face Transformers, scikit-learn, NumPy, pandas, Ollama